D01 | XOR decision block

All-case aligned observations: 38,224 | Train: 26,580 | Held-out test: 11,644

Test accuracy 0.627 | Majority baseline 0.621 | Balanced accuracy 0.514 | Macro-F1 0.424

B1 -> M02 | aligned support 14,501

B2 -> D02 | aligned support 23,723

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 923.5 AND c.AppType != "Limit raise" -> B2 [train n=16,031]

c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 923.5 AND c.AppType = "Limit raise" -> B2 [train n=2,890]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 923.5 AND e.FirstWithdrawal <= 1985.5 -> B2 [train n=1,641]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 923.5 AND e.FirstWithdrawal > 1985.5 -> B2 [train n=4,822]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10375 AND e.MonthlyCost <= 314.505 -> B1 [train n=597]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10375 AND e.MonthlyCost > 314.505 -> B1 [train n=250]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10375 AND e.FirstWithdrawal <= 25606.725 -> B2 [train
n=261]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10375 AND e.FirstWithdrawal > 25606.725 -> B2 [train n=88]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D02 | XOR decision block

All-case aligned observations: 23,723 | Train: 16,491 | Held-out test: 7,232

Test accuracy 0.908 | Majority baseline 0.908 | Balanced accuracy 0.500 | Macro-F1 0.476

B1 -> W_Call incomplete files | aligned support 2,135

B2 -> parallelGateway | aligned support 21,588

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 597.5 AND e.FirstWithdrawal <= 20203.655 -> B2 [train
n=5,477]

c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 597.5 AND e.FirstWithdrawal > 20203.655 -> B2 [train n=465]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 597.5 AND e.FirstWithdrawal <= 3000.5 -> B2 [train n=3,180]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 597.5 AND e.FirstWithdrawal > 3000.5 -> B2 [train n=6,827]

c.LoanGoal = "Remaining debt home" AND c.RequestedAmount <= 28115 AND c.RequestedAmount <= 12250 -> B2 [train n=144]

c.LoanGoal = "Remaining debt home" AND c.RequestedAmount <= 28115 AND c.RequestedAmount > 12250 -> B2 [train n=221]

c.LoanGoal = "Remaining debt home" AND c.RequestedAmount > 28115 AND e.OfferedAmount <= 31750 -> B2 [train n=51]

c.LoanGoal = "Remaining debt home" AND c.RequestedAmount > 28115 AND e.OfferedAmount > 31750 -> B2 [train n=126]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D03 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.837 | Majority baseline 0.651 | Balanced accuracy 0.767 | Macro-F1 0.792

B1 -> A_Submitted | aligned support 20,423
B2 -> M03 | aligned support 11,086

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "Limit raise" AND c.RequestedAmount <= 4500 -> B2 [train n=1,624]

c.AppType != "Limit raise" AND c.RequestedAmount > 4500 AND c.LoanGoal != "Unknown" -> B1 [train n=17,919]

c.AppType != "Limit raise" AND c.RequestedAmount > 4500 AND c.LoanGoal = "Unknown" -> B2 [train n=142]

c.AppType = "Limit raise" -> B2 [train n=2,371]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D04 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.788 | Majority baseline 0.695 | Balanced accuracy 0.831 | Macro-F1 0.778

B1 -> A_Cancelled | aligned support 9,921
B2 -> parallelGateway | aligned support 21,588

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 296 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 153.465 -> B1 [train n=3,105]

e.CreditScore <= 296 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 153.465 -> B1 [train n=7,610]

e.CreditScore <= 296 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 75 -> B2 [train n=1,501]

e.CreditScore <= 296 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 75 -> B2 [train n=539]

e.CreditScore > 296 -> B2 [train n=9,301]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D05 | XOR decision block

All-case aligned observations: 20,423 | Train: 14,272 | Held-out test: 6,151

Test accuracy 0.829 | Majority baseline 0.829 | Balanced accuracy 0.500 | Macro-F1 0.453

B1 -> W_Handle leads | aligned support 3,493

B2 -> M04 | aligned support 16,930

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount <= 9550 AND c.LoanGoal != "Home improvement" -> B2 [train
n=3,129]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount <= 9550 AND c.LoanGoal = "Home improvement" -> B2 [train
n=1,065]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount > 9550 AND c.LoanGoal != "Not speficied" -> B2 [train
n=6,942]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount > 9550 AND c.LoanGoal = "Not speficied" -> B2 [train
n=460]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 31727 AND c.RequestedAmount <= 20950 -> B2 [train
n=1,582]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 31727 AND c.RequestedAmount > 20950 -> B2 [train n=544]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 31727 AND c.RequestedAmount <= 52750 -> B2 [train n=410]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 31727 AND c.RequestedAmount > 52750 -> B2 [train n=140]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D06 | XOR decision block

All-case aligned observations: 36,857 | Train: 25,808 | Held-out test: 11,049

Test accuracy 0.856 | Majority baseline 0.856 | Balanced accuracy 0.500 | Macro-F1 0.461

B1 -> M05 | aligned support 5,348

B2 -> parallelGateway | aligned support 31,509

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 296 AND c.AppType != "Limit raise" AND e.OfferedAmount <= 9650 -> B2 [train n=4,252]

e.CreditScore <= 296 AND c.AppType != "Limit raise" AND e.OfferedAmount > 9650 -> B2 [train n=9,810]

e.CreditScore <= 296 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 225 -> B2 [train n=1,657]

e.CreditScore <= 296 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 225 -> B2 [train n=550]

e.CreditScore > 296 AND c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 853.5 -> B2 [train n=2,786]

e.CreditScore > 296 AND c.LoanGoal != "Remaining debt home" AND e.CreditScore > 853.5 -> B2 [train n=6,441]

e.CreditScore > 296 AND c.LoanGoal = "Remaining debt home" AND e.NumberOfTerms <= 168 -> B2 [train n=239]

e.CreditScore > 296 AND c.LoanGoal = "Remaining debt home" AND e.NumberOfTerms > 168 -> B2 [train n=73]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D07 | XOR decision block

All-case aligned observations: 21,588 | Train: 15,018 | Held-out test: 6,570

Test accuracy 0.877 | Majority baseline 0.826 | Balanced accuracy 0.904 | Macro-F1 0.824

B1 -> O_Accepted | aligned support 17,873
B2 -> A_Denied | aligned support 3,715

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 296 AND c.AppType != "New credit" AND e.NumberOfTerms <= 104.5 -> B1 [train n=622]

e.CreditScore <= 296 AND c.AppType != "New credit" AND e.NumberOfTerms > 104.5 -> B1 [train n=965]

e.CreditScore <= 296 AND c.AppType = "New credit" AND e.OfferedAmount <= 13450 -> B2 [train n=1,921]

e.CreditScore <= 296 AND c.AppType = "New credit" AND e.OfferedAmount > 13450 -> B2 [train n=2,209]

e.CreditScore > 296 -> B1 [train n=9,301]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D08 | XOR decision block

All-case aligned observations: 22,413 | Train: 15,582 | Held-out test: 6,831

Test accuracy 0.659 | Majority baseline 0.652 | Balanced accuracy 0.520 | Macro-F1 0.450

B1 -> M06 | aligned support 14,542
B2 -> M07 | aligned support 7,871

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 854.5 AND c.AppType != "Limit raise" -> B1 [train n=7,080]

c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 854.5 AND c.AppType = "Limit raise" -> B1 [train n=1,099]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 854.5 AND c.RequestedAmount <= 21800 -> B1 [train n=4,867]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 854.5 AND c.RequestedAmount > 21800 -> B1 [train n=1,660]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 5800 AND e.NumberOfTerms <= 114.5 -> B2 [train n=181]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 5800 AND e.NumberOfTerms > 114.5 -> B2 [train n=395]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 5800 AND e.CreditScore <= 723.5 -> B2 [train n=124]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 5800 AND e.CreditScore > 723.5 -> B1 [train n=176]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D09 | XOR decision block

All-case aligned observations: 21,588 | Train: 15,018 | Held-out test: 6,570

Test accuracy 0.686 | Majority baseline 0.677 | Balanced accuracy 0.577 | Macro-F1 0.575

B1 -> M06 | aligned support 7,046
B2 -> M07 | aligned support 14,542

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 593 AND c.AppType != "New credit" AND c.LoanGoal != "Existing loan takeover" -> B1 [train n=1,457]

e.CreditScore <= 593 AND c.AppType != "New credit" AND c.LoanGoal = "Existing loan takeover" -> B2 [train n=130]

e.CreditScore <= 593 AND c.AppType = "New credit" AND e.OfferedAmount <= 7750 -> B1 [train n=1,048]

e.CreditScore <= 593 AND c.AppType = "New credit" AND e.OfferedAmount > 7750 -> B2 [train n=3,083]

e.CreditScore > 593 AND e.FirstWithdrawal <= 1.5 AND c.LoanGoal != "Remaining debt home" -> B2 [train n=1,689]

e.CreditScore > 593 AND e.FirstWithdrawal <= 1.5 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=133]

e.CreditScore > 593 AND e.FirstWithdrawal > 1.5 AND e.NumberOfTerms <= 123.5 -> B2 [train n=6,983]

e.CreditScore > 593 AND e.FirstWithdrawal > 1.5 AND e.NumberOfTerms > 123.5 -> B2 [train n=495]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D10 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.772 | Majority baseline 0.539 | Balanced accuracy 0.772 | Macro-F1 0.771

B1 -> M09 | aligned support 16,738
B2 -> M12 | aligned support 14,771

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 593 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 154.965 -> B2 [train n=3,106]

e.CreditScore <= 593 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 154.965 -> B2 [train n=7,610]

e.CreditScore <= 593 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 375 -> B1 [train n=1,505]

e.CreditScore <= 593 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 375 -> B1 [train n=535]

e.CreditScore > 593 AND e.NumberOfTerms <= 60.5 AND e.OfferedAmount <= 5050 -> B1 [train n=798]

e.CreditScore > 593 AND e.NumberOfTerms <= 60.5 AND e.OfferedAmount > 5050 -> B1 [train n=2,975]

e.CreditScore > 593 AND e.NumberOfTerms > 60.5 AND c.AppType != "Limit raise" -> B1 [train n=5,240]

e.CreditScore > 593 AND e.NumberOfTerms > 60.5 AND c.AppType = "Limit raise" -> B1 [train n=287]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D11 | XOR decision block

All-case aligned observations: 18,934 | Train: 13,351 | Held-out test: 5,583

Test accuracy 0.781 | Majority baseline 0.781 | Balanced accuracy 0.500 | Macro-F1 0.438

B1 -> M09 | aligned support 14,771
B2 -> M12 | aligned support 4,163

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "Limit raise" AND e.OfferedAmount <= 6150 AND c.LoanGoal != "Not speficied" -> B1 [train n=2,056]

c.AppType != "Limit raise" AND e.OfferedAmount <= 6150 AND c.LoanGoal = "Not speficied" -> B1 [train n=168]

c.AppType != "Limit raise" AND e.OfferedAmount > 6150 AND e.NumberOfTerms <= 56.5 -> B1 [train n=2,341]

c.AppType != "Limit raise" AND e.OfferedAmount > 6150 AND e.NumberOfTerms > 56.5 -> B1 [train n=8,064]

c.AppType = "Limit raise" AND e.FirstWithdrawal <= 2200 AND e.OfferedAmount <= 7700 -> B1 [train n=58]

c.AppType = "Limit raise" AND e.FirstWithdrawal <= 2200 AND e.OfferedAmount > 7700 -> B1 [train n=603]

c.AppType = "Limit raise" AND e.FirstWithdrawal > 2200 -> B1 [train n=61]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D12 | XOR decision block

All-case aligned observations: 41,428 | Train: 29,010 | Held-out test: 12,418

Test accuracy 0.890 | Majority baseline 0.890 | Balanced accuracy 0.500 | Macro-F1 0.471

B1 -> M10 | aligned support 4,571
B2 -> M13 | aligned support 36,857

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 20.825 -> B2 [train n=4,973]

e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 20.825 -> B2 [train n=11,837]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 1508 -> B2 [train n=1,820]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 1508 -> B2 [train n=442]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount <= 2650 -> B2 [train n=89]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount > 2650 -> B2 [train n=249]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" -> B2 [train n=9,266]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" -> B2 [train n=334]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D13 | XOR decision block

All-case aligned observations: 39,988 | Train: 28,008 | Held-out test: 11,980

Test accuracy 0.922 | Majority baseline 0.922 | Balanced accuracy 0.500 | Macro-F1 0.480

B1 -> M13 | aligned support 3,131
B2 -> D14 | aligned support 36,857

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 605 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 0.32499999 -> B2 [train n=4,637]

e.CreditScore <= 605 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 0.32499999 -> B2 [train n=11,272]

e.CreditScore <= 605 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 13550 -> B2 [train n=530]

e.CreditScore <= 605 AND c.AppType = "Limit raise" AND e.OfferedAmount > 13550 -> B2 [train n=1,694]

e.CreditScore > 605 AND e.FirstWithdrawal <= 9692.8848 AND c.LoanGoal != "Remaining debt home" -> B2 [train n=5,848]

e.CreditScore > 605 AND e.FirstWithdrawal <= 9692.8848 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=206]

e.CreditScore > 605 AND e.FirstWithdrawal > 9692.8848 AND c.RequestedAmount <= 14100 -> B2 [train n=1,317]

e.CreditScore > 605 AND e.FirstWithdrawal > 9692.8848 AND c.RequestedAmount > 14100 -> B2 [train n=2,504]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D14 | XOR decision block

All-case aligned observations: 36,857 | Train: 25,808 | Held-out test: 11,049

Test accuracy 0.542 | Majority baseline 0.509 | Balanced accuracy 0.544 | Macro-F1 0.535

B1 -> W_Complete application | aligned support 18,944
B2 -> M11 | aligned support 17,913

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.RequestedAmount <= 4500 AND c.AppType != "Limit raise" AND e.OfferedAmount <= 10650 -> B1 [train n=565]

c.RequestedAmount <= 4500 AND c.AppType != "Limit raise" AND e.OfferedAmount > 10650 -> B1 [train n=1,360]

c.RequestedAmount <= 4500 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 12250 -> B1 [train n=113]

c.RequestedAmount <= 4500 AND c.AppType = "Limit raise" AND e.OfferedAmount > 12250 -> B1 [train n=446]

c.RequestedAmount > 4500 AND e.FirstWithdrawal <= 10801.5 AND e.FirstWithdrawal <= 194 -> B1 [train n=6,904]

c.RequestedAmount > 4500 AND e.FirstWithdrawal <= 10801.5 AND e.FirstWithdrawal > 194 -> B2 [train n=9,865]

c.RequestedAmount > 4500 AND e.FirstWithdrawal > 10801.5 AND c.RequestedAmount <= 10972.5 -> B2 [train n=140]

c.RequestedAmount > 4500 AND e.FirstWithdrawal > 10801.5 AND c.RequestedAmount > 10972.5 -> B2 [train n=6,415]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D15 | XOR decision block

All-case aligned observations: 21,588 | Train: 15,018 | Held-out test: 6,570

Test accuracy 0.618 | Majority baseline 0.494 | Balanced accuracy 0.619 | Macro-F1 0.616

B1 -> M16 | aligned support 10,779
B2 -> M15 | aligned support 10,809

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 852.5 AND e.OfferedAmount <= 26050 AND c.AppType != "Limit raise" -> B1 [train n=5,424]

e.CreditScore <= 852.5 AND e.OfferedAmount <= 26050 AND c.AppType = "Limit raise" -> B2 [train n=1,184]

e.CreditScore <= 852.5 AND e.OfferedAmount > 26050 AND e.CreditScore <= 730.5 -> B1 [train n=1,398]

e.CreditScore <= 852.5 AND e.OfferedAmount > 26050 AND e.CreditScore > 730.5 -> B1 [train n=455]

e.CreditScore > 852.5 AND e.OfferedAmount <= 25050 AND c.LoanGoal != "Remaining debt home" -> B2 [train n=5,157]

e.CreditScore > 852.5 AND e.OfferedAmount <= 25050 AND c.LoanGoal = "Remaining debt home" -> B1 [train n=138]

e.CreditScore > 852.5 AND e.OfferedAmount > 25050 AND e.FirstWithdrawal <= 2836 -> B1 [train n=447]

e.CreditScore > 852.5 AND e.OfferedAmount > 25050 AND e.FirstWithdrawal > 2836 -> B1 [train n=815]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.

D16 | XOR decision block

All-case aligned observations: 15,682 | Train: 11,020 | Held-out test: 4,662

Test accuracy 0.698 | Majority baseline 0.696 | Balanced accuracy 0.519 | Macro-F1 0.465

B1 -> M16 | aligned support 4,903
B2 -> M15 | aligned support 10,779

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 882.5 AND c.AppType != "Limit raise" -> B2 [train n=6,381]

c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 882.5 AND c.AppType = "Limit raise" -> B2 [train n=1,253]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 882.5 AND e.OfferedAmount <= 25800 -> B2 [train n=1,926]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 882.5 AND e.OfferedAmount > 25800 -> B2 [train n=822]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10125 AND e.CreditScore <= 720 -> B1 [train n=167]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10125 AND e.CreditScore > 720 -> B1 [train n=279]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10125 AND e.OfferedAmount <= 26000 -> B2 [train n=132]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10125 AND e.OfferedAmount > 26000 -> B2 [train n=60]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.45% is measured without these guards.